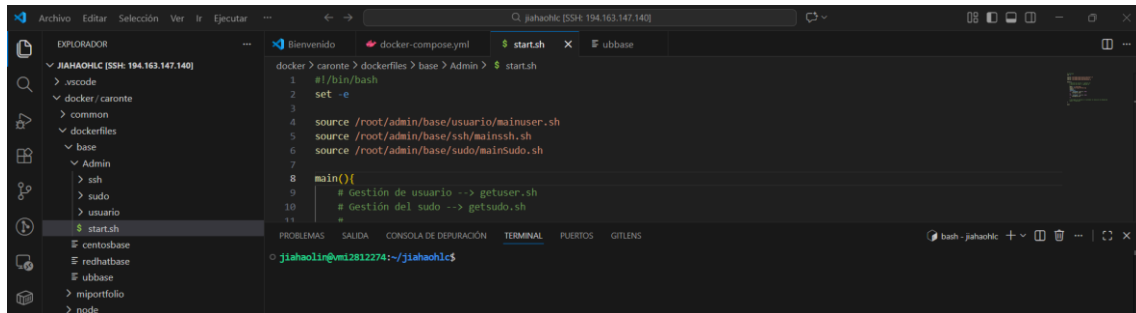


HELM

Es un packet manager de Kubernetes

Ahora vamos a crear una automatización con helm hacemos conexión remota a la VPS ya que vamos a trabajar directamente desde una VPS

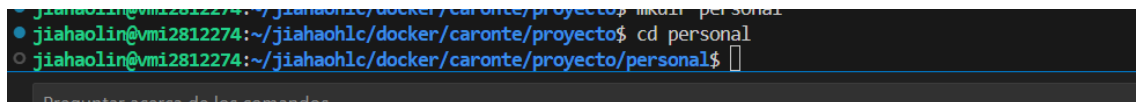


The screenshot shows a VS Code editor with a file explorer on the left displaying a directory structure for a project named 'jiahao'. The main editor area shows a Dockerfile with the following content:

```
1 #!/bin/bash
2 set -e
3
4 source /root/admin/base/usuario/mainuser.sh
5 source /root/admin/base/ssh/mainssh.sh
6 source /root/admin/base/sudo/mainsudo.sh
7
8 main(){
9     # Gestión de usuario --> getuser.sh
10    # Gestión del sudo --> getsudo.sh
11}
```

Below the editor, a terminal window is open, showing the command `startsh` being executed. The terminal output shows the user `jiahao` at the machine `vmi2812274` in the directory `~/jiahao`.

Entramos en proyecto y creamos personal

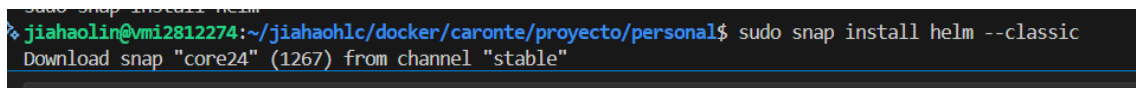


The screenshot shows a terminal window with the following commands and output:

```
jiahao@vmi2812274:~/jiahao/docker/caronte/proyecto$ mkdir personal
jiahao@vmi2812274:~/jiahao/docker/caronte/proyecto$ cd personal
jiahao@vmi2812274:~/jiahao/docker/caronte/proyecto/personal$
```

Y ponemos el siguiente comando

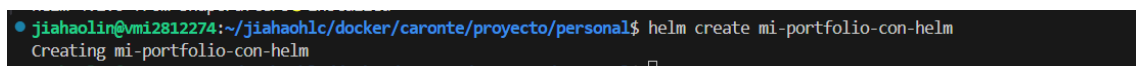
Para instalar Helm



The screenshot shows a terminal window with the following command and output:

```
jiahao@vmi2812274:~/jiahao/docker/caronte/proyecto/personal$ sudo snap install helm --classic
Download snap "core24" (1267) from channel "stable"
```

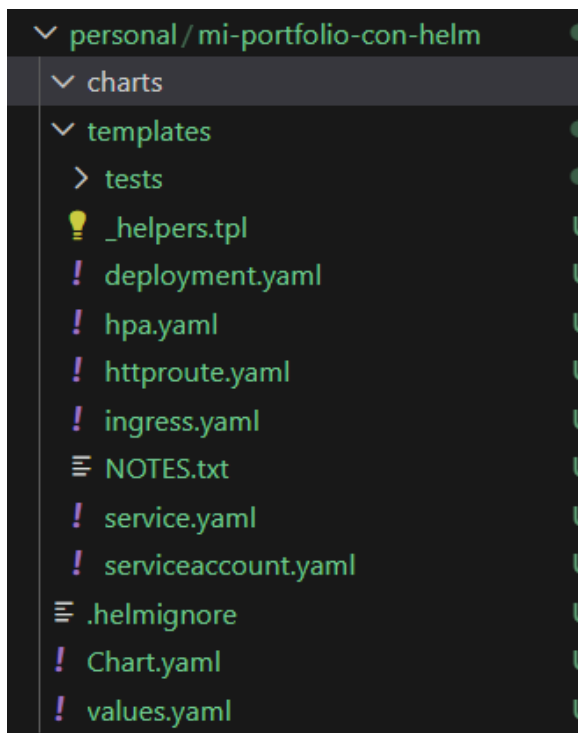
Y creamos un proyecto con portfolio



The screenshot shows a terminal window with the following command and output:

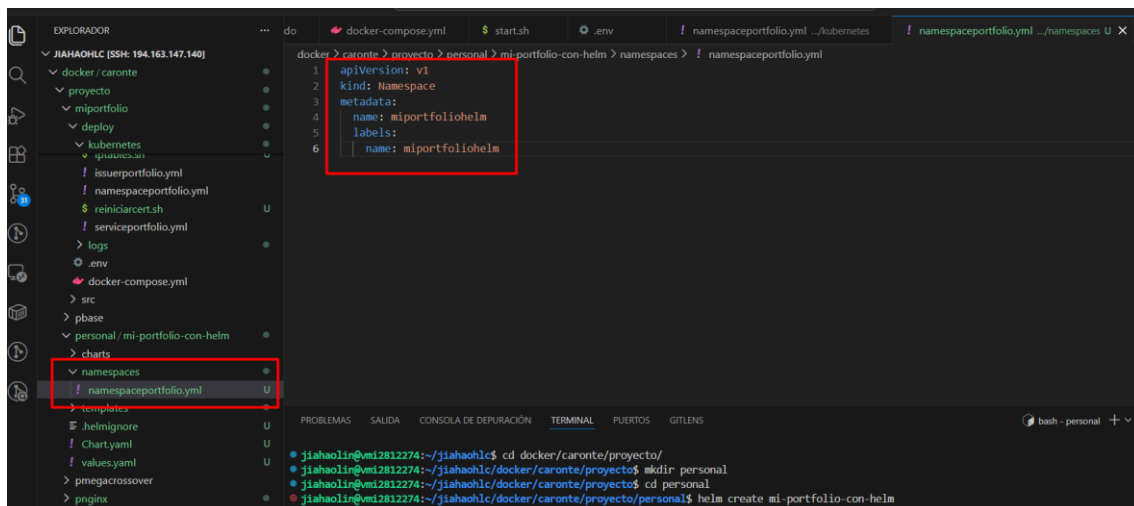
```
jiahao@vmi2812274:~/jiahao/docker/caronte/proyecto/personal$ helm create mi-portfolio-con-helm
Creating mi-portfolio-con-helm
```

Nos crea este listado de directorio

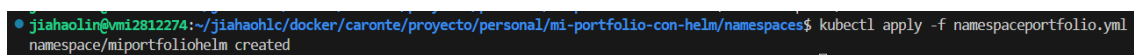


Estos servicios controlan nuestras aplicaciones.

Ahora lo que haremos es crear un namespaces.



Y lo aplicamos



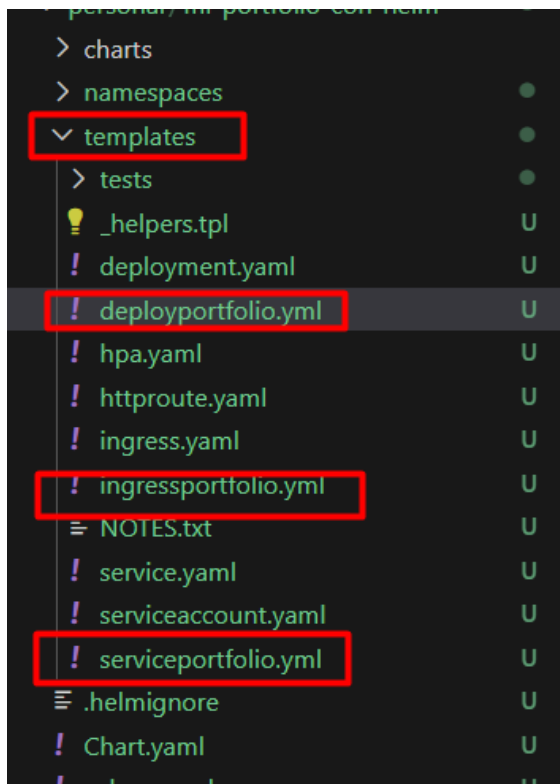
```

• jiahaolin@vm12812274:~/jiahaolc/docker/caronte/projecto/personal/mi-portfolio-con-helm/namespaces$ kubectl get namespace
NAME                STATUS   AGE
cert-manager        Active   6d
default              Active   13d
ingress              Active   13d
ingress-nginx        Active   5d23h
kube-node-lease      Active   13d
kube-public          Active   13d
kube-system          Active   13d
miportfoliohelm      Active   33s
projectoportfolio10  Active   6d
projectotaller       Active   6d23h
• jiahaolin@vm12812274:~/jiahaolc/docker/caronte/projecto/personal/mi-portfolio-con-helm/namespaces$

```

Ahora pasamos el deploy, el ingres y service

En template



Y en los archivos cambiamos el namespace a lo que hemos creado

En deploy

```
docker > caronte > proyecto > personal > mi-portfolio-con-helm > templates > ! deployportfolio.yml
1  apiVersion: apps/v1
2  kind: Deployment
3  metadata:
4    name: webportfolio-deployment
5    namespace: miportfoliohelm
6  spec:
7    replicas: 3
8    selector:
9      matchLabels:
10       app: webportfolio
11    template:
12      metadata:
13        labels:
14          app: webportfolio
15      spec:
16        containers:
17          - name: webportfolio
18            image: jiahaolin69/miportfolio
19            ports:
```

En ingress

```
docker > caronte > proyecto > personal > mi-portfolio-con-helm > templates > ! ingressportfolio.yml
1  apiVersion: networking.k8s.io/v1
2  kind: Ingress
3  metadata:
4    name: webtaller-ingress
5    namespace: miportfoliohelm
6  annotations:
7    nginx.ingress.kubernetes.io/rewrite-target: /
8  spec:
9    rules:
10     - host: portfolio.megacrossover.es
11       http:
12         paths:
13           - path: /
14             pathType: Prefix
15             backend:
16               service:
17                 name: portfolio-taller
18                 port:
19                   number: 80
```

Y en Service

```

docker > caronte > proyecto > personal > mi-portfolio-con-helm > templates > ! ser
1  apiVersion: v1
2  kind: Service
3  metadata:
4    name: portfolio-taller
5    namespace: miportfoliohelm
6  spec:
7    type: NodePort
8    selector:
9      app: webportfolio
10   ports:
11     - port: 80
12       targetPort: 80
13       nodePort: 30081

```

Y borramos lo anterior con

kubectl delete namespace miportfolio

```

jiahaojin@vmi2812274:~/jiahaojin/docker/caronte/proyecto/personal/mi-portfolio-con-helm/namespaces$ kubectl delete namespace proyectoportfoli
namespace "proyectoportfolio" deleted
[]

```

Ahora actualizamos el Helm con el siguiente comando

Helm upgrade --install [nombre del proyecto] . (indica el directorio) -n [namespace del proyecto] --create-namespace (no lo usamos nunca porque lo hemos creado en un fichero para tenerlo registrado)

```

projectcaronte: ~$ helm upgrade --install mi-portfolio-con-helm . -n miportfoliohelm
Release "mi-portfolio-con-helm" does not exist. Installing it now.
NAME: mi-portfolio-con-helm
LAST DEPLOYED: Tue Jan 27 13:21:42 2026
NAMESPACE: miportfoliohelm
STATUS: deployed
REVISION: 1
DESCRIPTION: Install complete
NOTES:
1. Get the application URL by running these commands:
  export POD_NAME=$(kubectl get pods --namespace miportfoliohelm -l "app.kubernetes.io/name=mi-portfolio-con-helm,app.kubernetes.io/instance=mi-portfolio-con-helm" -o jsonpath="{.items[0].metadata.name}")
  export CONTAINER_PORT=$(kubectl get pod --namespace miportfoliohelm $POD_NAME -o jsonpath="{.spec.containers[0].ports[0].containerPort}")
  echo "Visit http://127.0.0.1:8080 to use your application"
  kubectl --namespace miportfoliohelm port-forward $POD_NAME 8080:$CONTAINER_PORT

```

Si ejecutamos con `kubectl get all -n miportfoliohelm`

Y podemos observar que todo está corriendo

```
● jiahaolin@vm12812274:~/jiahaohlc/docker/caronte/proyecto/personal/mi-portfolio-con-helm$ kubectl get all -n miportfoliohelm
NAME                                READY    STATUS    RESTARTS   AGE
pod/mi-portfolio-con-helm-5654448b6-6sfmv    1/1      Running   0           6m26s
pod/webportfolio-deployment-9cb45fd7-52l8s    1/1      Running   0           6m26s
pod/webportfolio-deployment-9cb45fd7-bmjzn    1/1      Running   0           6m26s
pod/webportfolio-deployment-9cb45fd7-wpq5l    1/1      Running   0           6m26s

NAME                                TYPE          CLUSTER-IP    EXTERNAL-IP    PORT(S)          AGE
service/mi-portfolio-con-helm        ClusterIP      10.152.183.188 <none>         80/TCP           6m26s
service/portfolio-taller              NodePort       10.152.183.163 <none>         80:30081/TCP     6m26s

NAME                                READY    UP-TO-DATE    AVAILABLE    AGE
deployment.apps/mi-portfolio-con-helm  1/1      1              1            6m26s
deployment.apps/webportfolio-deployment  3/3      3              3            6m26s

NAME                                DESIRED    CURRENT    READY    AGE
replicaset.apps/mi-portfolio-con-helm-5654448b6    1          1          1        6m26s
replicaset.apps/webportfolio-deployment-9cb45fd7    3          3          3        6m26s
```

Y podemos observar que todo está corriendo

Si da error tenemos que hacer un `kubectl log` o describe con el nombre del POD y podemos ver el error que nos falla